

Conversational SHL Assessment Recommender

Design Choices

- FastAPI exposes a stateless GET /health and POST /chat; every request is reconstructed from the supplied message history.
- The scraped SHL Individual Test Solutions catalog in data/catalog.json is the source of truth. Pre-packaged Job Solutions are excluded by crawling only catalog type=1.
- The app builds an in-memory deterministic retriever at startup. It never scrapes during request handling.
- LLM generation is optional. Without keys, deterministic templates still clarify, recommend, refine, compare, and refuse.
- Final output is validated through Pydantic plus catalog-only URL repair. Recommendations can only be records from the local catalog.

Retrieval Setup

- Indexed fields: name, description, raw detail text, test type, job levels, languages, duration, remote testing, adaptive/IRT, and keywords.
- Lexical retrieval combines word TF-IDF with character n-grams so terms like Java, C++, .NET, OPQ, GSA, and product variants still match.
- Optional semantic retrieval is supported if sentence-transformers is installed, but the deployable baseline stays lightweight.
- Rule boosts cover exact skills, assessment families, role/seniority context, requested test types, and constraints such as no personality or short tests.
- A transparent intent boost map promotes real catalog records for sales, contact center, graduate cognitive, leadership, and OPQ/personality queries; final items are still selected from data/catalog.json.
- Software relevance penalties demote unrelated K tests for Java/Python/backend queries while keeping exact skill tests at the top.
- Broad intents return up to 10 items for Recall@10; narrow skill requests usually return 5.

Prompt Design

- The LLM receives only a conversation summary and retrieved catalog evidence; it is not allowed to invent names, URLs, test types, durations, or capabilities.
- Deterministic code extracts constraints, retrieves candidates, validates final recommendations, and handles refusals before any LLM text is trusted.
- Clarification asks one high-value question only when there is no role, skill, assessment family, job description, or test-type clue.
- Comparison responses are grounded in catalog fields and explicitly say when duration, remote testing, adaptive/IRT, or job-level metadata is missing.

Evaluation Method

- scripts/run_eval.py measures schema compliance, catalog validity, Recall@10, lexical relevance, approximate groundedness, and behavior probe pass rate.
- Behavior probes cover vague clarification, recommendation, public traces, refinement, comparison, off-topic refusal, prompt-injection refusal, URL hallucination, and max-10 limits.
- Recall@10 uses labeled expected catalog names/URLs where available; relevance and groundedness use local catalog evidence, not model self-judgment.

What Did Not Work

- Pure LLM recommendations hallucinated assessment names and URLs, so the model was removed from the source-of-truth path.
- Keyword-only retrieval missed synonyms and job-description phrasing; adding character n-grams and rule boosts improved technical and acronym matching.
- Asking several clarification questions hurt the eight-turn evaluator cap; one concise question worked better.
- Long prompts increased latency and did not improve recall enough to justify using them on every request.

How Improvement Was Measured

- Baseline keyword retrieval was compared with hybrid lexical retrieval plus rule boosts on Java, Python, sales, contact center, graduate, leadership, OPQ, and Excel cases.
- Before intent boosts and public traces, Recall@10 was 0.717 with schema/catalog/behavior all 1.000. After the intent boost map, broad-query sizing, software penalties, and 10 public traces, Recall@10 is 1.000 with schema compliance 1.000, catalog validity 1.000, groundedness 1.000, and behavior pass rate 1.000.

- LLM used: configurable through LLM_PROVIDER and MODEL_NAME; deployment defaults to Gemini 2.5 Flash when GEMINI_API_KEY is set.
- AI tools used: Codex for implementation assistance, with catalog checks, tests, and evaluation used as review gates.